

		thus displacement is increasing at an increasing rate.
5	B	Option A: check gradient of tangent with the steepest slope. Option B: check gradient of tangent with the gentlest slope and with result of Option A, 20 m s^{-1} is possible as it is between the max. and min. speed. Option C: average speed for the first 200 m is $200/15 = 13.3 \text{ m s}^{-1}$. Option D: for the same time interval of 5 s, the distance travelled is less for 20 s to 25 s, thus the average speed is smaller.
6	A	When D is below maximum value of f , there is no net force and a is 0